

Lecture 2

The potential outcome framework

Outline

- Definition of concepts: intervention, potential outcomes, causal effects
- The SUTVA assumption
- Causal estimand and assignment mechanism
- Perfect doctor example
- An alternative language: Causal directed acyclic graph (DAG)
- Suggested reading: Imbens and Rubin Chapter 1.1-1.8, 1.10, 3.3 Peng's book Chapter 1.3, 2

Causal questions require specified interventions

Causality concerns an action applied to a unit, with an outcome assessed at a specified time.

“No causation without manipulation” (Rubin, 1975)

A hypothetical intervention need not be performed, but its meaning must be coherent.

- HDL: a particular drug regimen, compared with a specified alternative.
- Smoking: stopping versus continuing during a specified period.
- Vaccination: two doses versus placebo, with symptomatic COVID-19 as the endpoint.
- Schooling: an additional year of education and later earnings.

Treatment, unit, and potential outcomes

- **Treatment:** An active intervention applied at a particular moment in time, whose effects we wish to assess relative to no intervention (the control)
 - Take aspirin now versus do not take it now. The comparison can also be between two active treatments.
- **Unit:** A physical object at a particular moment in time
 - One person now, assess headache two hours later under either treatment.
- **Potential outcomes:** Outcomes that would be observed if active treatment is applied and that would be observed if control treatment is applied
 - The headache outcomes that would occur at the same later time under the two alternatives.

Does a before–after change reveal the causal effect for this unit?

Unit	Potential Outcomes		Causal Effect
	$Y(\text{Aspirin})$	$Y(\text{No Aspirin})$	
You	No Headache	Headache	Improvement due to Aspirin

Individual causal effects

Treatment assignment: $W_i = 1$ if treated $W_i = 0$ if control

1. Headache gone only with aspirin:

$Y(\text{Aspirin}) = \text{No Headache}$, $Y(\text{No Aspirin}) = \text{Headache}$

Positive effect

2. No effect of aspirin, with a headache in both cases:

$Y(\text{Aspirin}) = \text{Headache}$, $Y(\text{No Aspirin}) = \text{Headache}$

No effect

3. No effect of aspirin, with the headache gone in both cases:

$Y(\text{Aspirin}) = \text{No Headache}$, $Y(\text{No Aspirin}) = \text{No Headache}$

No effect

4. Headache gone only without aspirin:

$Y(\text{Aspirin}) = \text{Headache}$, $Y(\text{No Aspirin}) = \text{No Headache}$

Negative effect

For the same unit:

$$\tau_i = Y_i(1) - Y_i(0)$$

The sign depends on outcome coding. Here, improvement means headache relief.

Covariates and the observed outcome

- **Pre-treatment covariates** are characteristics the current treatment assignment could not have affected.
- Examples: baseline headache severity, age, education, family background, and pre-training earnings.
- Pre-training earnings are a covariate. Post-training earnings are the outcome of interest.

Voters	Contact	Turnout		Age	Gender
i	W_i	$Y_i(1)$	$Y_i(0)$	X_{i1}	X_{i2}
1	1	1	?	20	M
2	0	?	0	55	F
3	0	?	1	40	F
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	1	0	?	62	M

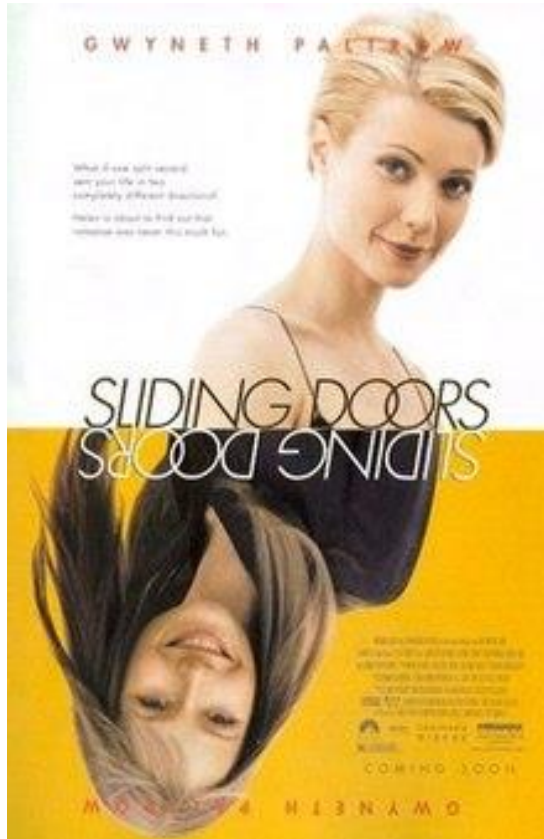
Baseline
covariates

Observed outcome:

$$Y_i = Y_i(W_i)$$

Only one potential outcome is observed.

Potential outcomes in a story



Sliding Doors (1998)

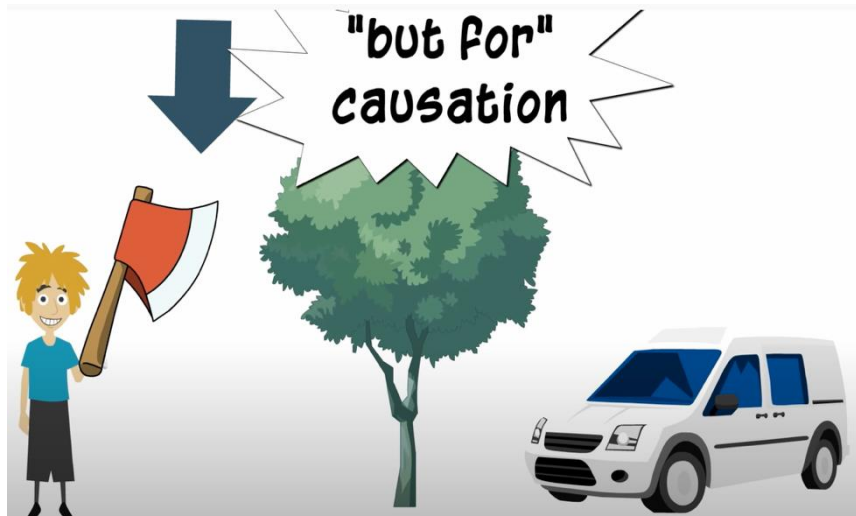
The film follows two possible paths for the same woman, depending on whether she catches a train.

$Y(1)$ The future if she catches the train

$Y(0)$ The future if she misses the train

In a real study, we observe only the path that occurs.

Counterfactual reasoning in the but-for test



The but-for test is a test commonly used in both tort law and criminal law to determine actual causation.

“But for the harmful event, what would have happened?”

Compare the person’s actual economic position with the position they would have occupied at the same time without the harmful event.

Which economic position can we observe, and which must we infer?

Causal contrasts and treatment levels

- Causal effect for unit i : $\tau_i = Y_i(1) - Y_i(0)$
- We can also use other quantities to define the causal effect:
ex. Log fold change $\log(Y_i(1)/Y_i(0))$, percentage change $\frac{Y_i(1) - Y_i(0)}{Y_i(0)} \times 100\%$.
- Any causal quantity is a function of potential outcomes
- Extension to Non-binary treatment:
 - Categorical: $Y_i(0), Y_i(1), \dots, Y_i(K - 1)$
 - Continuous: $Y_i(t)$ for any $t \in \mathbb{R}$

Quick check: when would a ratio or percentage change be undefined?

The fundamental missing-data problem

Voters	Contact	Turnout		Age	Gender
i	W_i	$Y_i(1)$	$Y_i(0)$	X_{i1}	X_{i2}
1	1	1	?	20	M
2	0	?	0	55	F
3	0	?	1	40	F
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	1	0	?	62	M

- Only one potential outcome is observed for each unit.
- For fixed units, condition on their potential outcomes. Assignment selects which outcome we observe.
- Under random assignment, observed outcomes vary with assignment. Sampling units can introduce another source of randomness.

Where does the randomness come from: the assignment, the sampled units, or both?

Underlying assumptions of the potential outcome framework: SUTVA

- Temporal ordering and the two SUTVA conditions
 1. **Causal ordering:** Y_i cannot causally affect W_i
 - Feedback over time (e.g. dosage adjustment) requires treatment histories
 - Pre-treatment symptoms may affect assignment without reversing causal ordering
 2. **No interference** between units : A unit's potential outcome does not change when other units' treatments change.

$$Y_i(W_1, \dots, W_n) = Y_i(W_i)$$

- Example: if we have no contact with each other, whether you take an aspirin has no effect on the status of my headache
 - **Counter-examples:**
 - vaccine effect, effect of job training programs (peer effects)
 - Spillovers: treatment or exposure of one unit changes another unit's outcome

Underlying assumptions of the potential outcome framework: SUTVA

3. No different forms or versions of each treatment level (consistency)

- By setting $W_i = 1$ for taking the aspirin, we assume that there is only one version of the tablets
 - May hide different doses, formulations, or administration
 - Specify relevant versions, or assume the versions grouped under a treatment label have the same outcome for that unit.
 - Consistency links the observed outcome to the potential outcome under the intervention received.
 - Different surgeons can be compatible with SUTVA if each unit's intervention is well-defined
- **SUTVA:** Stable Unit Treatment Value Assumption 2 + 3

Discuss:

Different aspirin doses; a vaccinated classmate; baseline symptoms driving treatment.

Which concern treatment versions, interference, or neither?

Immutable characteristics and specified interventions

- **Immutable characteristics:** sex, race, age, etc.
- Can immutable characteristics have meaningful causal effects?
- Examples:
 - **Race:** resume experiment
 - Fictitious résumés sent to job ads in Boston and Chicago received randomly assigned African-American- or White-sounding names. White-sounding names received about 50% more callbacks.

Bertrand & Mullainathan (2004), American Economic Review 94(4), 991–1013.

- **Gender:** village leadership experiment
- One third of village-council head positions in India were randomly reserved for women. Leaders invested more in infrastructure directly relevant to the needs of their own gender.
- It changes the eligibility rule and resulting leadership. It does not randomly change the sex of a fixed individual.

Chattopadhyay & Duflo (2004), Econometrica 72(5), 1409–1443.

Both studies show how a broad question about an immutable characteristic can be turned into a question about a specified intervention.

Race as a treatment: What changes, and what stays fixed?

Harvard admissions data for the classes of 2014–2019

Arcidiacono's model estimated that Asian-American applicants' admission rate would rise by about **1 percentage point (19%)** if they were **treated as White**.

What does “treated as White” mean?

- **Same submitted materials, different perceived race?**
Should Harvard's evaluation—including its personal rating—then be allowed to change?
- **Different racial experiences throughout life?**
Background, opportunities, and the submitted application might also change.

A concrete disagreement: Harvard's personal rating

- **Card hold it fixed:** it may capture relevant personal qualities.
- **Arcidiacono did not hold it fixed:** it may already reflect discrimination.

For immutable characteristics, defining the causal effect requires specifying what changes—and what stays fixed.

Average causal effects: sample and population

A **causal estimand** specifies the causal quantity and the units over which it is defined.

In the sample

All sample units

$$\text{SATE} = \frac{1}{n} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\}$$

Treated sample units

$$\text{SATT} = \frac{1}{n_1} \sum_{i=1}^n W_i \{Y_i(1) - Y_i(0)\}$$
$$n_1 = \sum_{i=1}^n W_i$$

Choose the target:

Training effect among this sample's trainees, or among all eligible workers in the population?

In a population

All population units

$$\text{PATE} = \mathbb{E}(Y_i(1) - Y_i(0))$$

Treated population units

$$\text{PATT} = \mathbb{E}(Y_i(1) - Y_i(0) \mid W_i = 1)$$

Other causal estimands

Conditional average treatment effect

$$\tau(\mathbf{x}) = \mathbb{E}(Y_i(1) - Y_i(0) \mid \mathbf{X}_i = \mathbf{x})$$

Averages among units with baseline covariates $\mathbf{X} = \mathbf{x}$.

Quantile of individual effects

$$Q_{Y(1)-Y(0)}(\alpha)$$

Difference of marginal quantiles

$$Q_{Y(1)}(\alpha) - Q_{Y(0)}(\alpha)$$

These generally differ. The first depends on the joint distribution of the two potential outcomes.

Odds ratio for a binary outcome

$$\frac{P(Y_i(1) = 1)/P(Y_i(1) = 0)}{P(Y_i(0) = 1)/P(Y_i(0) = 0)}$$

Quantiles refer to the chosen percentile; the 50th percentile is the median.

Observed comparisons and treatment assignment

The **assignment mechanism** governs which treatment each unit receives.

Unit	Unknown			Known	
	Potential Outcomes		Causal Effect	Actual	Observed
	$Y(\text{Aspirin})$	$Y(\text{No Aspirin})$		Treatment W_i	Outcome Y_i^{obs}
You	No Headache	Headache	Improvement due to Aspirin	Aspirin	No Headache
I	No Headache	No Headache	None	No Aspirin	No Headache

Both patients have no headache. Does this imply that aspirin has no causal effect?

Next: could a doctor who always chooses the better treatment create a misleading comparison?

Perfect doctor: treatment choice and effects

Treatment 0 = drug; treatment 1 = surgery. Larger values mean longer survival.

The doctor chooses the better treatment for every patient.

Patient	Drug (0)	Surgery (1)
1	1	7
2	6	5
3	1	5
4	8	7

Pair work

- Which treatment does each patient receive?
- What is the average surgery-minus-drug causal effect?
- What is the observed surgery-minus-drug mean difference?

Perfect doctor: why the comparison reverses

Full potential outcomes

Unit	Potential Outcomes		Causal Effect
	$Y_i(0)$	$Y_i(1)$	$Y_i(1) - Y_i(0)$
Patient #1	1	7	6
Patient #2	6	5	-1
Patient #3	1	5	4
Patient #4	8	7	-1
Average	4	6	2

SATE: $(6 - 1 + 4 - 1) / 4 = +2$

Observed data after assignment

Unit	Treatment	Observed Outcome
i	W_i	Y_i^{obs}
Patient #1	1	7
Patient #2	0	6
Patient #3	1	5
Patient #4	0	8

Surgery mean = 6; drug mean = 7

Observed difference = -1

Why? The causal effect compares treatments for the same units.
The observed contrast compares different, selected groups.

What does the group-mean imputation assume?

Observed potential outcomes

Unit	W_i	$Y_i(0)$	$Y_i(1)$
1	1	?	7
2	0	6	?
3	1	?	5
4	0	8	?

Missing values filled by the other group's mean

Unit	W_i	$Y_i(0)$	$Y_i(1)$
1	1	7	7
2	0	6	6
3	1	7	5
4	0	8	6

Check patient 3

Would a perfect doctor choose surgery if the drug outcome were 7 and surgery gave 5?

The unchosen outcome must be no greater than the chosen outcome. This restricts missing values but does not determine them.

Experimental and observational assignment

Randomized experiment

The researcher specifies a known assignment distribution:

$$P(W|X)$$

- Assignment may be completely random or depend on baseline covariates, as in a stratified experiment.
- Examples include laboratory, survey, and field experiments.

Observational study

- Treatments may be real interventions chosen by patients, clinicians, or institutions. The researcher observes their assignment.
- Knowledge of treatment rules and recorded covariates can help, but does not ensure all confounders are measured.

Internal validity concerns the study target. External validity concerns generalizing beyond it.

History of the potential-outcome framework

Neyman (1923)

Defined estimands in randomized experiments as functions of potential outcomes.

Fisher (1925)

Developed randomization-based reasoning for assessing causal effects in experiments.

Rubin (1974–1978)

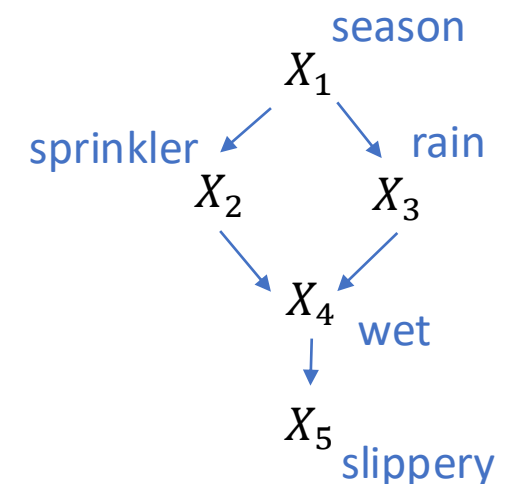
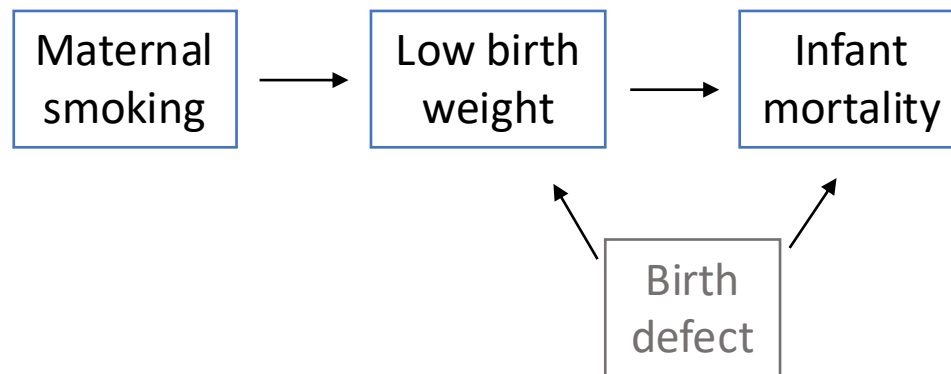
Extended the potential-outcome framework to observational settings and emphasized the assignment mechanism.

Causal directed acyclic graphs

- This course focuses on using the potential outcome language, but there is another language to quantitatively describe causal inference
- Causal DAG: a directed graph with no cycles and the arrows has a causal meaning
- Nodes are random variables (may not be observed)
- PA_j : parents of X_j (nodes that have direct arrows to X_j)

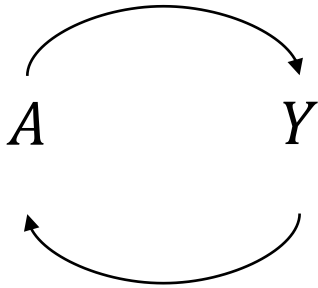
$PA \rightarrow X$
parent descendant

Examples:

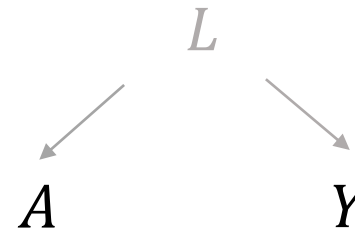


Assumptions behind a causal graph

- Acyclicity rules out a directed path returning to its starting node.
- The common causes of any pair of variables in the graph must be in the graph (either observed or unobserved)



A directed cycle violates acyclicity



An unobserved common cause

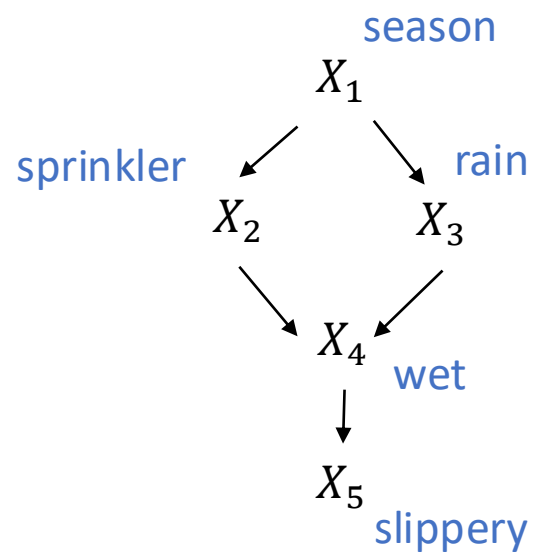
The intervention and unit definitions also require care about treatment versions and interference.

A formal intervention sets a node. Its meaning and feasibility are substantive questions.

Structural equations and potential outcomes

Structural equation model

$$E_{X_j} \perp PA_j$$



Error-parent independence
is a modeling assumption.

$$X_4 = f_4(X_2, X_3, E_{X_4})$$

Each node has an unknown deterministic function and an error, held fixed across interventions: causal relationship from ALL parents to a node

Joint intervention on the parents represented by potential outcome

$$X_j(pa_j) = f_j(pa_j, E_{X_j})$$

Set both sprinkler and rain:

$$X_4(x_2, x_3) = f_4(x_2, x_3, E_{X_4})$$

Set sprinkler alone:

$$X_4(x_2) = f_4(x_2, X_3(x_2), E_{X_4}) = f_4(x_2, X_3, E_{X_4})$$

Comparison between the two languages

- One reference:

Imbens, G. W. (2020). Potential outcome and directed acyclic graph approaches to causality: Relevance for empirical practice in economics. *Journal of Economic Literature*, 58(4), 1129-79.

- DAGs display causal relationships and clarify assumptions about pathways and common causes
- Causal discovery from data requires additional assumptions and may not identify a unique DAG
- A bare graph does not specify effect heterogeneity. Structural functions can represent it
- Potential outcomes state estimands and assignment assumptions explicitly
- Both languages can describe interventions and experiments, with suitable assumptions

Reference papers to read

Causal effects of immutable Characteristics:

- Bertrand, M., & Mullainathan, S. (2004). Are Emily and Greg more employable than Lakisha and Jamal? A field experiment on labor market discrimination. *American economic review*, 94(4), 991-1013.
<https://www.aeaweb.org/articles?id=10.1257/0002828042002561>
- Chattopadhyay, R., & Duflo, E. (2004). Women as policy makers: Evidence from a randomized policy experiment in India. *Econometrica*, 72(5), 1409-1443.
https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1468-0262.2004.00539.x?casa_token=1SVxHRxC5hQAAAAA:fz62pJ2QMeg41laiRMIERS5K1psbjVovA3-fgD5OTPjb2C-UkLuiQvv0CV2HSFq9z7r3WxxmH_p7NqE
- Arcidiacono, P., Kinsler, J., & Ransom, T. (2022). Asian American discrimination in Harvard admissions. *European Economic Review*, 104079. <https://www.iza.org/publications/dp/13172/asian-american-discrimination-in-harvard-admissions>

Lord's paradox:

- Holland, P. W., & Rubin, D. B. (1983). On Lord's paradox. *Principals of modern psychological measurement*, 3-25.
<https://onlinelibrary.wiley.com/doi/abs/10.1002/j.2333-8504.1982.tb01321.x>

Comparison between DAG and potential outcome framework:

- Imbens, G. W. (2020). Potential outcome and directed acyclic graph approaches to causality: Relevance for empirical practice in economics. *Journal of Economic Literature*, 58(4), 1129-79.
<https://www.aeaweb.org/articles?id=10.1257/jel.20191597>